

STAT 6410

Homework

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1. Chapter 2, Problem 2.1

Let Ω_i , $i = 1, 2$ be two nonempty sets and $T : \Omega_1 \rightarrow \Omega_2$ be a map. Then for any collection $\{A_\alpha : \alpha \in I\}$ of subsets of Ω_2 , show that:

$$T^{-1}\left(\bigcup_{\alpha \in I} A_\alpha\right) = \bigcup_{\alpha \in I} T^{-1}(A_\alpha)$$

$$\text{and } T^{-1}\left(\bigcap_{\alpha \in I} A_\alpha\right) = \bigcap_{\alpha \in I} T^{-1}(A_\alpha)$$

Further, $(T^{-1}(A))^c = T^{-1}(A^c)$ for all $A \subset \Omega_2$. (These are known using *de Morgan's* laws.)

Ans:

Proof. Part 1: WTS: $T^{-1}\left(\bigcup_{\alpha \in I} A_\alpha\right) = \bigcup_{\alpha \in I} T^{-1}(A_\alpha)$

Part 1.a: $T^{-1}\left(\bigcup_{\alpha \in I} A_\alpha\right) \subset \bigcup_{\alpha \in I} T^{-1}(A_\alpha)$

Let $\omega \in T^{-1}\left(\bigcup_{\alpha \in I} A_\alpha\right)$ be given.

Thus, $T(\omega) \in \bigcup_{\alpha \in I} A_\alpha \implies \exists \beta \in I$ s.t. $T(\omega) \in A_\beta$ and $\omega \in T^{-1}(A_\beta)$.

$T^{-1}(A_\beta) \subset \bigcup_{\alpha \in I} T^{-1}(A_\alpha) \implies \omega \in \bigcup_{\alpha \in I} T^{-1}(A_\alpha)$

Thus, $T^{-1}\left(\bigcup_{\alpha \in I} A_\alpha\right) \subset \bigcup_{\alpha \in I} T^{-1}(A_\alpha)$

Part 1.b: $T^{-1}\left(\bigcup_{\alpha \in I} A_\alpha\right) \supset \bigcup_{\alpha \in I} T^{-1}(A_\alpha)$

Let $\omega \in \bigcup_{\alpha \in I} T^{-1}(A_\alpha)$ be given.

Thus, $\exists \beta \in I$ s.t. $T(\omega) \in A_\beta$ and $\omega \in T^{-1}(A_\beta)$.

Since, $A_\beta \subset \bigcup_{\alpha \in I} A_\alpha$, $\omega \in T^{-1}\left(\bigcup_{\alpha \in I} A_\alpha\right)$

Thus, $T^{-1}\left(\bigcup_{\alpha \in I} A_\alpha\right) = \bigcup_{\alpha \in I} T^{-1}(A_\alpha)$.

Part 2: WTS: $T^{-1}\left(\bigcap_{\alpha \in I} A_\alpha\right) = \bigcap_{\alpha \in I} T^{-1}(A_\alpha)$

Part 2.a: $T^{-1}\left(\bigcap_{\alpha \in I} A_\alpha\right) \subset \bigcap_{\alpha \in I} T^{-1}(A_\alpha)$

Let $\omega \in T^{-1}\left(\bigcap_{\alpha \in I} A_\alpha\right)$ be given. Thus, $T(\omega) \in \bigcap_{\alpha \in I} A_\alpha$.

$\implies \forall \alpha \in I, T(\omega) \in A_\alpha. \therefore \omega \in T^{-1}(A_\alpha) \forall \alpha \in I \implies \omega \in \bigcap_{\alpha \in I} T^{-1}(A_\alpha)$

Part 2.b: $T^{-1}(\bigcap_{\alpha \in I} A_\alpha) \supset \bigcap_{\alpha \in I} T^{-1}(A_\alpha)$

Let $\omega \in \bigcap_{\alpha \in I} T^{-1}(A_\alpha)$ be given. Thus, $\forall \alpha \in I, \omega \in T^{-1}(A_\alpha)$ and $T(\omega) \in A_\alpha$.
 $T(\omega) \in \bigcap_{\alpha \in I} A_\alpha \implies \omega \in T^{-1}(\bigcap_{\alpha \in I} A_\alpha)$

Thus, $T^{-1}(\bigcap_{\alpha \in I} A_\alpha) = \bigcap_{\alpha \in I} T^{-1}(A_\alpha)$.

Part 3: WTS: $(T^{-1}(A))^c = T^{-1}(A^c)$ for all $A \subset \Omega_2$

Let $A \subset \Omega_2$ be given.

Part 3.a: WTS $(T^{-1}(A))^c \subset T^{-1}(A^c)$

Let $\omega \in (T^{-1}(A))^c$ be given.

Thus, $\omega \notin T^{-1}(A) \implies T(\omega) \notin A \implies T(\omega) \in A^c$
 $\therefore \omega \in T^{-1}(A^c)$

Part 3.b: WTS $(T^{-1}(A))^c \supset T^{-1}(A^c)$

Let $\omega \in T^{-1}(A^c)$ be given.

Thus, $T(\omega) \in A^c \implies T(\omega) \notin A$.

$\implies \omega \notin T^{-1}(A) \implies \omega \in (T^{-1}(A))^c$

Thus, $(T^{-1}(A))^c = T^{-1}(A^c)$ for all $A \subset \Omega_2$ □

2. Chapter 2, Problem 2.3

Let $f, g : \Omega \rightarrow \mathbb{R}$ be $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}) \rangle$ -measurable. Set

$$h(\omega) = \frac{f(\omega)}{g(\omega)} I(g(\omega) \neq 0), \omega \in \Omega.$$

Verify that h is $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}) \rangle$ -measurable.

Ans:

Proof. $x \equiv (x_1, x_2) \in \mathbb{R}^2$.

Define: $h_1(x) \equiv \frac{x_1}{x_2} I(x_2 \neq 0)$.

$h_1^{-1}((-\infty, a]) \equiv \{(x_1, x_2) \in \mathbb{R}^2 : x_2 = 0, a \leq 0\}$

$\cup \{(x_1, x_2) \in \mathbb{R}^2 : x_2 > 0, x_1 - ax_2 \leq 0\} \cup \{(x_1, x_2) \in \mathbb{R}^2 : x_2 < 0, x_1 - ax_2 \geq 0\}$

$h_1^{-1}((-\infty, a]) \in \mathcal{B}(\mathbb{R}^2)$ for all $a \in \mathbb{R}$. Thus, h_1 is $\langle \mathcal{B}(\mathbb{R}^2), \mathcal{B}(\mathbb{R}) \rangle$ -measurable, by Definition 2.1.2.

By Proposition 2.1.3, $\eta = (f, g)$ is $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}^2) \rangle$ -measurable.

By Proposition 2.1.1, $h = \nu \circ h_1$ is $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}) \rangle$ -measurable. □

3. Chapter 2, Problem 2.6

Let X_i , $i = 1, 2, 3$ be random variables on a probability space $(\Omega, \mathcal{F}, P)$. Consider the random equation (in $t \in \mathbb{R}$):

$$X_1(\omega)t^2 + X_2(\omega)t + X_3(\omega) = 0. \quad (6.1)$$

- (a) Show that $A \equiv \{\omega \in \Omega : \text{Equation (6.1) has two distinct real roots}\} \in \mathcal{F}$.
(b) Let $T_1(\omega)$ and $T_2(\omega)$ denote the two roots of (6.1) on A . Let

$$f_i(\omega) = \begin{cases} T_i(\omega) & \text{on } A \\ 0 & \text{on } A^c \end{cases},$$

$i = 1, 2$. Show that (f_1, f_2) is $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}^2) \rangle$ -measurable.

Ans:

- (a) *Proof.* We can rewrite (without changing the set composition) A as:
 $A = \{\omega \in \Omega : X_2(\omega)^2 - 4X_1(\omega)X_3(\omega) > 0\} \cap \{\omega \in \Omega : X_1(\omega) \neq 0\}$. This pulls from the definition of the determinant of a quadratic equation from elementary algebra.

Because X_1, X_2, X_3 are random variables, they are $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}) \rangle$ -measurable transformations from Ω to $\mathbb{R}$.

By Corollary 2.1.4., since X_1, X_2, X_3 are $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}) \rangle$ -measurable functions from Ω to $\mathbb{R}$, this collection is closed under addition, multiplication, and scalar multiplication. Thus, $X_2^2 - 4X_1X_3$ is $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}) \rangle$ -measurable (and a random variable on the probability space $(\Omega, \mathcal{F}, P)$).

Thus, by the definition of a random variable, $A \in \mathcal{F}$. □

- (b) *Proof.* Again, referring to elementary algebra, we can define T_1 and T_2 as:

$$T_1(\omega) = \frac{-X_2(\omega) - \sqrt{X_2^2(\omega) - 4X_1(\omega)X_3(\omega)}}{2X_1(\omega)}$$

$$T_2(\omega) = \frac{-X_2(\omega) + \sqrt{X_2^2(\omega) - 4X_1(\omega)X_3(\omega)}}{2X_1(\omega)}$$

Define f_i :

$$f_i(\omega) = T_i(\omega)I(\omega \in A) = T_i(\omega)I(X_1(\omega) \neq 0)I(X_2^2(\omega) - 4X_1(\omega)X_3(\omega) > 0)$$

$$g(\omega) \equiv \sqrt{X_2^2(\omega) - 4X_1(\omega)X_3(\omega)}I(X_2^2(\omega) - 4X_1(\omega)X_3(\omega) > 0)$$

$g : \Omega \rightarrow \mathbb{R}$ is piecewise continuous on Ω . Thus, by problem 2.2, g is $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}) \rangle$ -measurable.

$h(\omega) = -X_2(\omega)$ and $l(\omega) = 2X_1(\omega)$ are both $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}) \rangle$ -measurable by Corollary 2.1.4.

By, Corollary 2.1.4, $h(\omega) - g(\omega)$ and $h(\omega) + g(\omega)$ are both $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}) \rangle$ -measurable.

Finally, by Problem 2.3,

$$f_1(\omega) = \frac{h(\omega) - g(\omega)}{l(\omega)} I(l(\omega) \neq 0) = T_2(\omega) I(\omega \in A) \text{ and}$$

$$f_2(\omega) = \frac{h(\omega) + g(\omega)}{l(\omega)} I(l(\omega) \neq 0) = T_1(\omega) I(\omega \in A)$$

are $\langle \mathcal{F}, \mathcal{B}(\mathbb{R}) \rangle$ -measurable. □

4. Extra Problem A

Let $X : \Omega_1 \rightarrow \Omega_2$ be a $\langle \mathcal{F}_1, \mathcal{F}_2 \rangle$ measurable transformation. Consider the collection (of subsets of Ω_1) defined by $\sigma\langle X \rangle = \{X^{-1}(B), B \in \mathcal{F}_2\}$. Then show that

- (a) $\sigma\langle X \rangle$ is σ -algebra ,
- (b) $\sigma\langle X \rangle$ is the *smallest* σ -algebra that makes X measurable, (i.e. if X is $\langle \mathcal{G}, \mathcal{F}_2 \rangle$ -measurable, then $\sigma\langle X \rangle \subset \mathcal{G}$).

Ans:

- (a) *Proof. Part 1:* WTS: $\Omega_1 \in \sigma\langle X \rangle$.

This is trivial.

Part 2: WTS: $A \in \sigma\langle X \rangle \implies A^c \in \sigma\langle X \rangle$

Let $A \in \sigma\langle X \rangle$ be given.

Thus, there exists some $B \in \mathcal{F}_2$ such that $A = X^{-1}(B)$.

$A^c = (X^{-1}(B))^c = X^{-1}(B^c)$, by Problem 2.1.

$B^c \in \mathcal{F}_2$ since $\mathcal{F}_2$ is a σ -algebra .

Thus, $A = X^{-1}(B^c) \in \sigma\langle X \rangle$

Part 3: WTS $A_1, A_2, \dots, A_n \in \sigma\langle X \rangle \implies \bigcup_{\alpha \in I} A_\alpha \in \sigma\langle X \rangle$, where $I = (1, 2, \dots, n)$ for some fixed n

Let $A_1, A_2, \dots, A_n \in \sigma\langle X \rangle$ be given. Thus there exists $B_i \in \mathcal{F}_2$ such that $A_\alpha = X^{-1}(B_\alpha)$ for all $\alpha \in I$.

$\bigcup_{\alpha \in I} A_\alpha = \bigcup_{\alpha \in I} X^{-1}(B_\alpha) = X^{-1}(\bigcup_{\alpha \in I} B_\alpha)$, by Problem 2.1

$\bigcup_{\alpha \in I} B_\alpha \in \mathcal{F}_2$ since $\mathcal{F}_2$ is a σ -algebra .

Thus, $\bigcup_{\alpha \in I} A_\alpha = X^{-1}(\bigcup_{\alpha \in I} B_\alpha) \in \sigma\langle X \rangle$.

Thus, $\sigma\langle X \rangle$ is a σ -algebra . $\square$

(b) *Proof.* Suppose X is $\langle \mathcal{G}, \mathcal{F}_2 \rangle$ -measurable.

Thus, for all $B \in \mathcal{F}_2$, $X^{-1}(B) \in \mathcal{G} \implies \sigma\langle X \rangle = \{X^{-1}(B), B \in \mathcal{F}_2\} \subset \mathcal{G}$.

Thus, $\sigma\langle X \rangle$ is the *smallest* σ -algebra that makes X measurable. $\square$

5. Extra Problem B

Give an example of a function f that is **not** $\langle \mathcal{B}(\mathbb{R}), \mathcal{B}(\mathbb{R}) \rangle$ -measurable.

Ans: From the results of Lebesgue Measure on $\mathbb{R}$, we know that there exists $A \in \mathbb{R}$ such that $B \notin \mathcal{B}(\mathbb{R})$.

$f : (\mathbb{R}, \mathcal{B}(\mathbb{R})) \rightarrow (\mathbb{R}, \mathcal{B}(\mathbb{R}))$

As such, $f(\omega) = \begin{cases} 1 & \omega \in B \\ 0 & \omega \notin B \end{cases}$ is not $\langle \mathcal{B}(\mathbb{R}), \mathcal{B}(\mathbb{R}) \rangle$ -measurable.

Indicator functions are only measurable if A is in $\mathcal{B}(\mathbb{R})$.